A Guideline for the Optimum Fin Width Considering Hot-Carrier and NBTI Degradation in MuGFETs

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***Abstract*—Considering both hot-carrier and negative-bias-temperature-instability (NBTI) degradations, a guideline for the optimum fin width in p-channel multiple-gate MOSFETs (p–MuGFETs) has been suggested. From the different dependences of the fin width on both hot-carrier and NBTI degradations in p-MuGFETs, the optimum fin width has been extracted empirically to maximize the device lifetime. The optimum fin width increases as the stress temperature increases. When the fin width is narrower than the optimum fin width, the device lifetime is dominantly determined by NBTI degradation. However, when the fin width is wider than the optimum fin width, the device lifetime is determined by hot-carrier degradation. Considering both hot-carrier and NBTI degradations, the tradeoff between fin width and fin number has been discussed.**

***Index Terms*—Hot carriers, multiple-gate MOSFET (MuGFET), negative-bias temperature instability (NBTI), silicon-on-insulator (SOI) technology.**

I. Introduction

As CMOS device scaling approaches toward the nanometer regime, the multiple-gate MOSFET (MuGFET) is considered as the most promising candidate for future nanometer electronic devices due to large current drive, excellent control of short-channel effects, and a fabrication process which is similar to that of standard CMOS. In bulk or silicon-on-insulator (SOI) MOSFETs, the current drive can be increased by simply increasing the width of the transistor. This is not an option in MuGFETs since varying the fin width has profound implications on the electrical properties of the device. In regard to electrostatic integrity of MuGFETs, the ratio of the minimum gate length to the minimum fin width should be larger than 1.5 times [1]. Therefore, it could be said that the minimum feature size in MuGFET technology is the fin width (W_F), not the gate length. For MuGFETs, the minimum effective fin width (W_{EFF}) can be expressed as $W_{\mathrm{EFF}} = n_F \times (2H_F + W_F)$. Here, n_F is the fin number, and H_F is the fin height. Since the drive current increases with W_{EFF}, one should increase n_F or H_F and W_F to increase the drive current. A fin aspect ratio of $H_F/W_F = 2$ is usually considered as a good tradeoff between current drive and manufacturability [1].

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At present levels of gate oxide thickness and electric fields, negative-bias-temperature-instability (NBTI) and hot-carrier effects are reported to be serious reliability problem in nanometer-scale p-channel MuGFET (p-MuGFET) technologies. NBTI degradation is more significant for narrow FinFETs [2]–[4], and hot-carrier degradation at elevated temperature is more significant in devices with wider fins [5], [6]. Several studies on the influence of W_F on the NBTI and hot-carrier degradations have been reported [2]–[6], but a study on the device design guidelines for determining the optimum W_F and n_F to maximize the device lifetime has not been investigated.

In this letter, NBTI- and hot-carrier-induced device degradations with different W_F's in p-MuGFETs with a nitride oxide buried layer (NOx-BL) as the buried insulator layer have been experimentally studied. The optimum W_F to maximize the device lifetime with stress temperature and the tradeoff between W_F and n_F are proposed.

II. Device Preparation and Measurement

Devices were fabricated on SOI wafers with a NOx-BL as the buried insulator layer. The thicknesses of the silicon film and the gate oxide are 60 and 1.8 nm, respectively. The SOI film is p-type with a boron doping concentration of approximately 2×10^{15} cm^{-3}. The thicknesses of BOX and Si_3N_4 are 70 and 30 nm, respectively. The thickness of the TiSiN gate is 6 nm, and the effective gate length is 65 nm. The drawn fin widths of the measured devices were in the range of 55–90 nm, but the final widths after processing were decreased in the range of 32–67 nm. The detailed fabrication process of MuGFETs used in this work has already been reported previously [7]. NBTI stress was applied with the gate electrode held at a constant negative bias of $V_{\mathrm{GS}} = -1.5$ V, and hot-carrier stress was applied at $V_{\mathrm{GS}} = V_{\mathrm{DS}} = -1.5$ V under temperature between 25 °C and 125 °C. For NBTI characterization, the on-the-fly method using an Agilent B1500A precision semiconductor parameter analyzer was applied to avoid measurement errors. I_{DSAT} was monitored at $V_{\mathrm{DS}} = -1.0$ V and $V_{\mathrm{GS}} = -1.4$ V.

III. Results and Discussion

Fig. 1 shows the hot-carrier- and NBTI-induced drain current degradations ($\Delta I_{\mathrm{DSAT}}/I_{\mathrm{DSAT}}$) as functions of the stress time for different values of W_F. The gate length and n_F are 95 nm and 20, respectively. The stress temperature is 75 °C. It can clearly be seen that hot-carrier and NBTI degradations have opposite dependences on W_F. In order to explain the more significant hot-carrier degradation in wider fins, the enhanced

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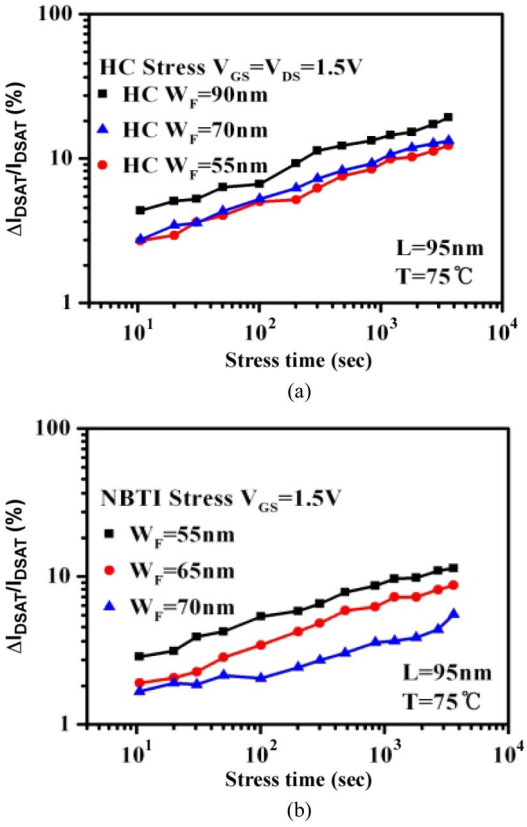

Fig. 1. (a) Hot-carrier- and (b) NBTI-induced drain current degradations as functions of the stress time for different W_F's.

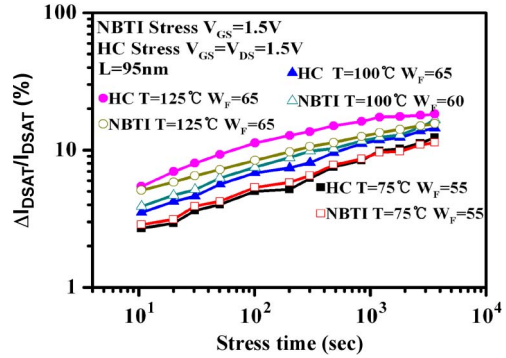

Fig. 2. Comparison of hot-carrier and NBTI degradations as functions of the stress time for different stress temperatures.

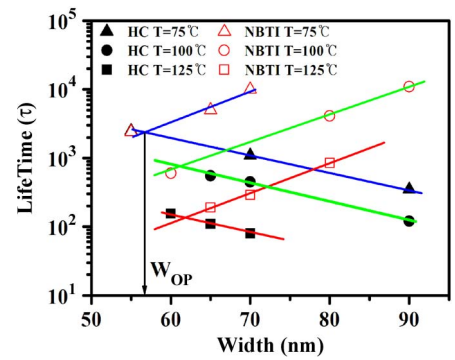

Fig. 3. Lifetime versus W_F for different stress temperatures.

impact ionization due to the nonlocal effects and reduced series resistance [5] and the enhancement of floating-body effects [2] have been suggested. Since the substrate is floated during the experiment and the positive charges in NOx-BL enhance the floating-body effects [8], we believe that the enhanced impact ionization rate and/or the parasitic bipolar transistor effects play a dominant role in the increased device degradation with the increase of W_F [9]. The reasons for the more significant NBTI degradation in narrow devices were explained by out-diffusion of hydrogen at the corners in multiple-gate devices [3].

To compare the dependences of W_F on both hot-carrier and NBTI degradations, $\Delta I_{\mathrm{DSAT}}/I_{\mathrm{DSAT}}$ in p-MuGFETs which have a similar W_F is plotted in Fig. 2 as a function of the stress time for different stress temperatures. We can easily see that $\Delta I_{\mathrm{DSAT}}/I_{\mathrm{DSAT}}$ of wider devices at high temperature is more significant under channel hot-carrier stress than under NBTI stress. This had been explained by the concurrent NBTI and hot-carrier effects at high stress temperature [10]–[13]. Although the dependence of H_F on the device degradation has not been explored here, it can be concluded that the concurrent effects are more significant in MuGFETs as the fin aspect ratio H_F/W_F decreases. From Fig. 2, the hot-carrier degradation of $W_F = 55$ nm is nearly the same as the NBTI degradation at a stress temperature of 75 °C. This indicates that NBTI degradation could be more significant than hot-carrier degradation depending on the stress temperature and W_F. Therefore, we can explore the dependence of W_F on the hot-carrier and NBTI degradations from the prediction of device lifetime with stress temperature. The device lifetime is defined as the stress time necessary to reach $\Delta I_{\mathrm{DSAT}}/I_{\mathrm{DSAT}} = 10\%$. The NBTI and hot-carrier lifetimes were estimated for different W_F's ranging from 55 to 90 nm and stress temperatures ranging from 25 °C to 125 °C.

Fig. 3 shows the device lifetime as a function of W_F with different stress temperatures. The optimum fin width (W_{OP}) to maximize the device lifetime considering both hot-carrier and NBTI degradations can be extracted from the crossing points at each stress temperature. The extracted W_{OP} is about 55 nm at 75 °C, and it increases as the stress temperature increases. When W_F is narrower than W_{OP}, the device lifetime is dominantly governed by NBTI degradation. However, when W_F is wider than W_{OP}, the device lifetime is dominantly governed by hot-carrier degradation. Therefore, Fig. 3 can be used as a design guideline for W_{OP} considering both hot-carrier and NBTI degradations in p-MuGFETs. Considering that H_F is 60 nm in this work, the optimum fin aspect ratio at 75 °C is less than $H_F/W_F = 1.0$ at the stress condition of $V_{\mathrm{GS}} = V_{\mathrm{DS}} = -1.5$ V. Although the fin aspect ratio of $H_F/W_F = 2$ is desirable in respect of manufacture, it is worth noting that the optimum fin aspect ratio depends on the gate length and operating temperature. When the stress bias voltage is constant, W_{OP} will be increased with the increase of the gate length because hot-carrier degradation is decreased with gate length, but NBTI degradation is independent of the gate length. Since the device lifetime depends on the measurement bias conditions, W_{OP} could be a function of the measurement bias voltages.

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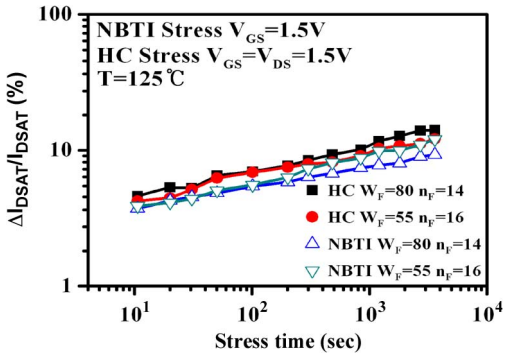

Fig. 4. For $W_{\mathrm{EFF}} = 2.8\ \mu$m, $\Delta I_{\mathrm{DSAT}}/I_{\mathrm{DSAT}}$ with different W_F's and n_F's under hot-carrier and NBTI stress at 125 °C.

When the effective total width is constant ($W_{\mathrm{EFF}} = 2.8\ \mu$m), $\Delta I_{\mathrm{DSAT}}/I_{\mathrm{DSAT}}$ of the devices with different W_F's and n_F's under channel hot-carrier and NBTI stress at 125 °C is plotted in Fig. 4. It is worth noting that the initially hot-carrier-induced $\Delta I_{\mathrm{DSAT}}/I_{\mathrm{DSAT}}$ of the devices with small W_F (large n_F) and large W_F (small n_F) is nearly the same, but eventually, the large W_F (small n_F) becomes more significant. However, the reverse is observed in NBTI degradation.

Eventually, the small W_F (large n_F) becomes more significant in NBTI degradation. The increased hot-carrier degradation for large W_F is due to the enhanced impact ionization rate and/or the enhanced floating-body effects, and the increased NBTI degradation for large n_F is due to the increased out-diffusion of hydrogen at the increased number of corners in MuGFETs. Therefore, the observed results indicate that W_F and n_F play a more dominant role in hot-carrier and NBTI degradations, respectively. When W_{EFF} is constant, it is better to reduce W_F and increase n_F to suppress hot-carrier degradation and to increase W_F and reduce n_F to suppress NBTI degradation. This observation can be used as a guideline for device design.

IV. Conclusion

The extraction of the optimum fin width has been suggested to maximize the device lifetime in MuGFETs. The optimum fin width increases as the stress temperature increases. When the fin width is narrower than the optimum fin width, the device lifetime is dominantly governed by NBTI degradation. However, when the fin width is wider than the optimum fin width, the device lifetime is governed by hot-carrier degradation. Although the fin aspect ratio of $H_F/W_F = 2$ is desirable in respect of manufacture, we observed that the optimum fin aspect ratio depends on both gate length and operating temperature.

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